(Multi-Agent) Reinforcement Learning: a primer, with applications in economics

Natural-gas storage modelling by deep reinforcement learning

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*All views and opinions are those of the speaker(s) and do not necessarily reflect the position of Bank of Italy

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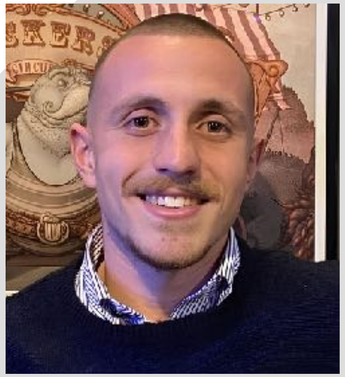
Acknowledgments first!

Joint work with



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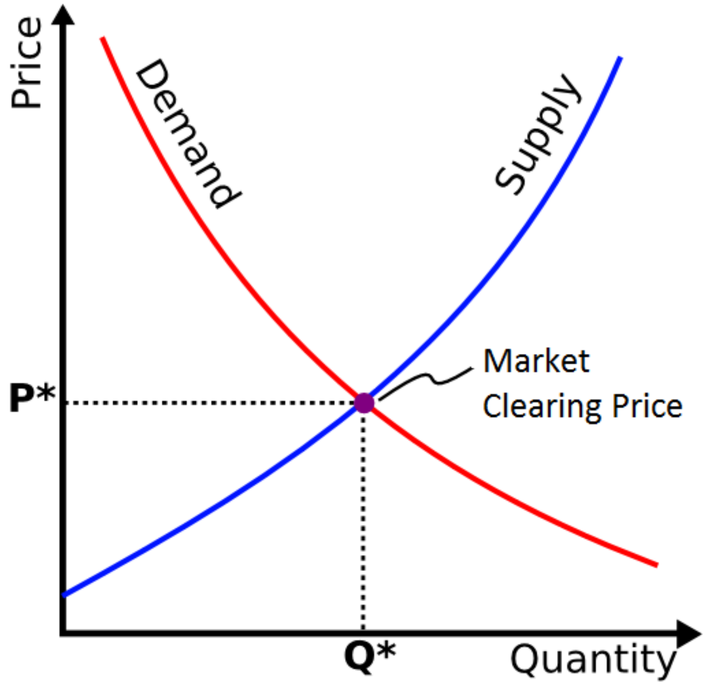
EU natural-gas storage requirement

- In **June 2022**, the EU required Member States that their underground natural gas storage facilities are **at least 90% full** by the 1st of November of each year

- The refilling target is **controversial**, and the debate is stil open: **complex assessment of the trade-offs** between market efficiency and energy security

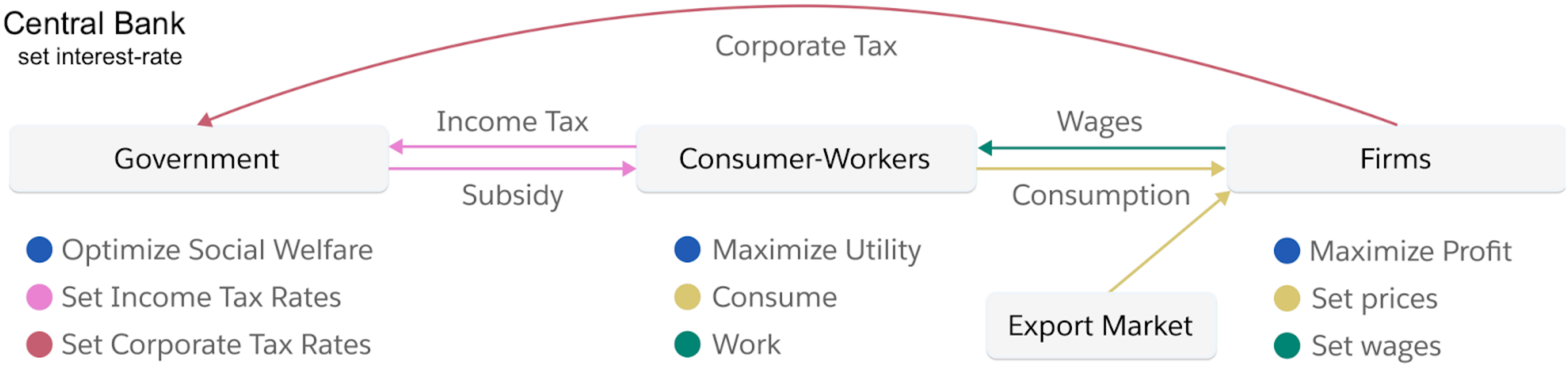
Market clearning in practice



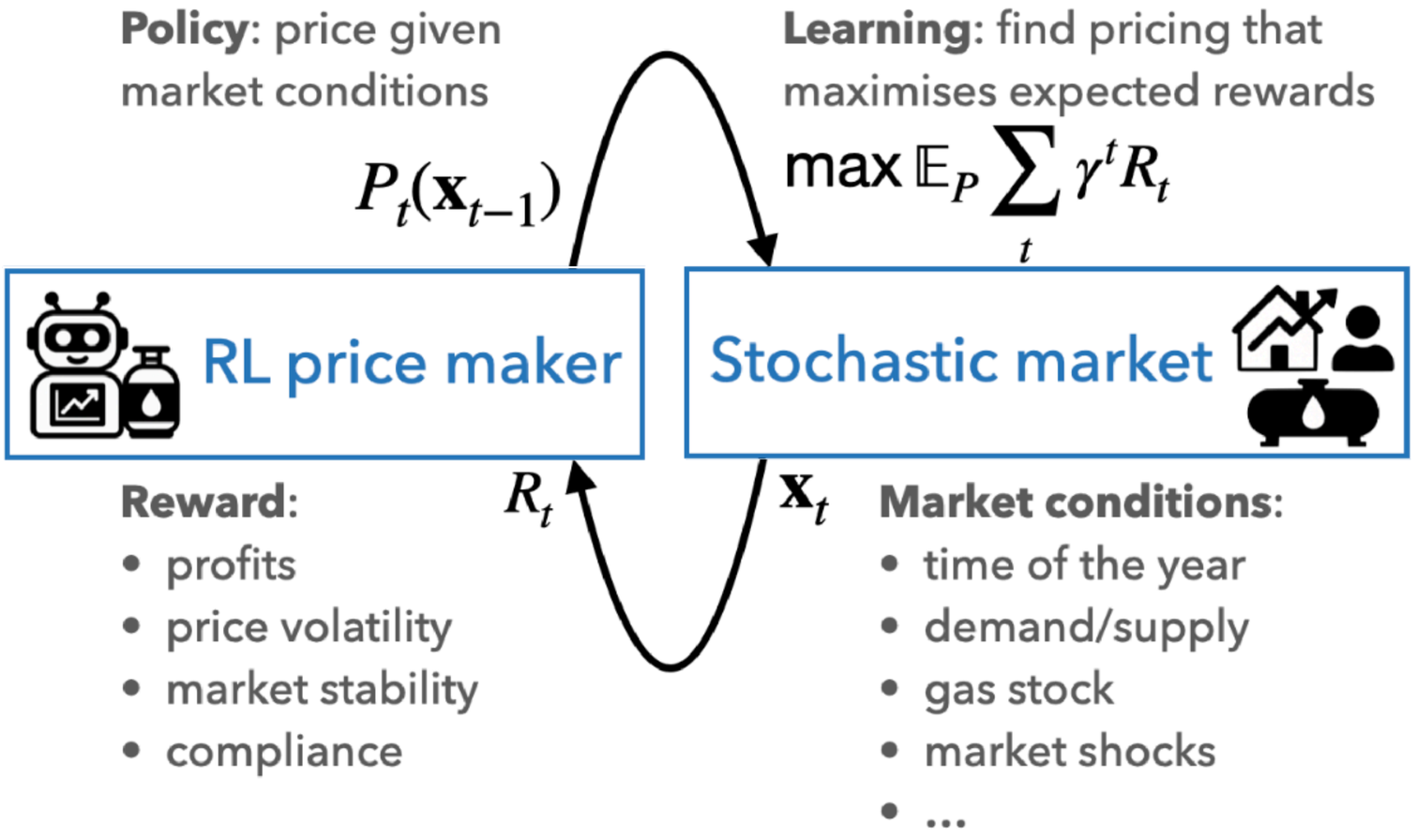
- In the traditional economic view, markets are assumed to be in/near "equilibrium": if price is above equilibrium, excess supply pushes it down; if price is below equilibrium, excess demand pushes it up.

- In practice, inefficiencies, rationing, queues, and other nonequilibrium phenomena are frequently observed.

AI agents in simulation models: A practical application



The GasRL simulator



- Storage operators **balance prices** by **injecting** gas during periods of low demand (e.g., summer) and **withdrawing** it during periods of high demand (e.g., winter)

- A monopolistic RL operator is a reasonable assumption for Italy as **SNAM owns 90%** of the natural-gas storage capabilities

The GasRL simulator

Stochastic market

1. Retail gas price

$$P_t$$

2. Demand/supply price signals (logs)

$$p_t^d = \ln\left[\lambda_d e^{p_{t-1}^d} + (1-\lambda_d)P_t\right]$$

$$p_t^s = \ln\left[\lambda_s e^{p_{t-1}^s} + (1-\lambda_s)P_t\right]$$

λ: stickiness parameters

3. Demand and supply (logs)

$$d_t = S_t - \eta_d p_t^d + u_t^d$$

$$s_t = \eta_s p_t^s + u_t^s$$

S: seasonal demand

u: AR(1) shocks

η: price elasticities

4. Excess demand

$$D_t = e^{d_t} - e^{s_t}$$

net gas witdrawals (can be negative)

- Parameters calibrated on the Italian market

5. Natural-gas stock

$$I_{t+1} = \begin{cases} I_t - D_t, & -(I_{\max} - I_t) \le D_t \le I_t \\ 0, & D_t > I_t \quad \text{[unmet demand]} \\ I_{\max}, & D_t < -(I_{\max} - I_t) \quad \text{[wasted supply]} \end{cases}$$

market can fail: stocks cannot go below zero or above maximum!

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The GasRL simulator

RL price maker

- Reward parameters
  - $\theta_v = 20$
  - $\theta_m = 1000$
  - $\theta_n = 0 / 750$
- Nine-dimensional observation space
- Time horizon of $T = 360$ steps (30 years x 12 months)

Bank account evolution

$$g_t = (1 + r)\, g_{t-1} - \tau I_{t-1} - P_t \Delta I_t$$

interests *costs* *sold/bought gas*

Reward

$$R_t = \underline{\Delta g_t} - \theta_v \underline{(\Delta p_t)}^2 - \theta_m \underline{m_t(1 + \tilde{m}_t)} - \theta_n \underline{n_t(1 + \tilde{n}_t)}$$

profits *price volatility* *market failure* *regulatory compliance*

Observation space

$$\mathbf{x}_t = (S_t, \cos(\phi_t), \sin(\phi_t), u_t^d, u_t^s, p_t^d, p_t^s, \ln(0.5 + I_t), p_t)$$

seasonal signals *AR(1) shocks* *past prices and inventory*

Policy optimisation

$$P_t(\mathbf{x}_{t-1}) = \arg\max_P \mathbb{E}_P \left[\sum_{t=1}^{T} \gamma^t R_t \right]$$

solved by state of the art RL schemes like PPO or SAC

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The GasRL simulator

Policy: price given market conditions

$P_t(\mathbf{x}_{t-1})$

Learning: find pricing that maximises expected rewards

$$\max \mathbb{E}_P \sum_t \gamma^t R_t$$

R_t

$\mathbf{x}_t$

Reward:

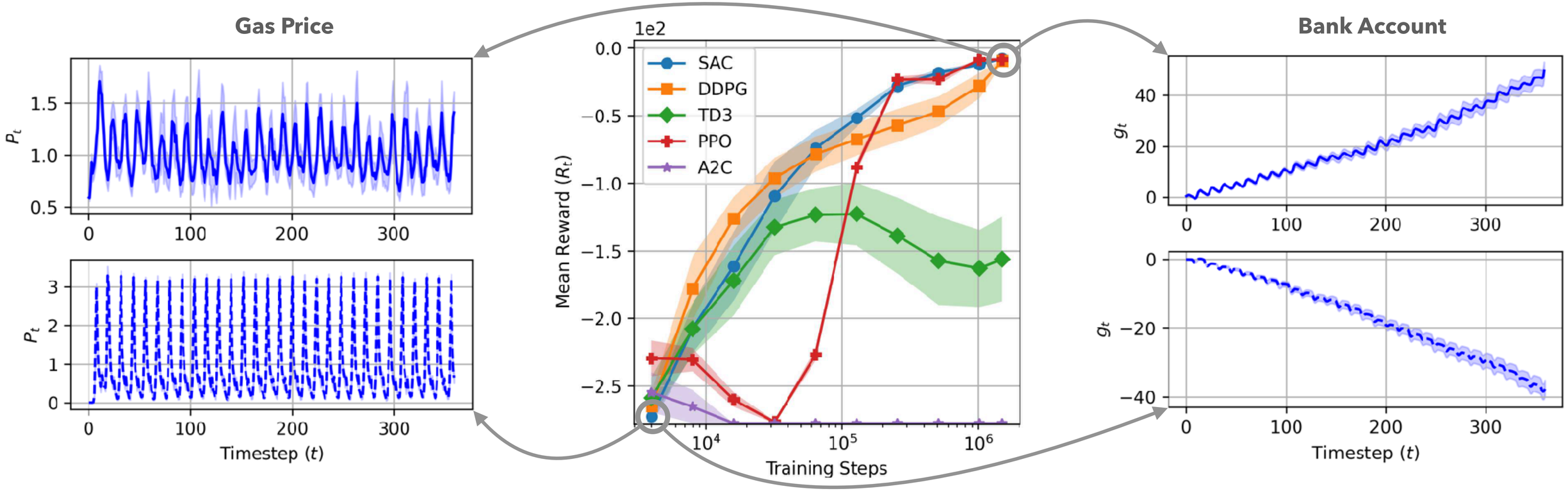
- profits
- price volatility
- market stability
- compliance

Market conditions:

- time of the year
- demand/supply
- gas stock
- market shocks
- ...

Does it work?

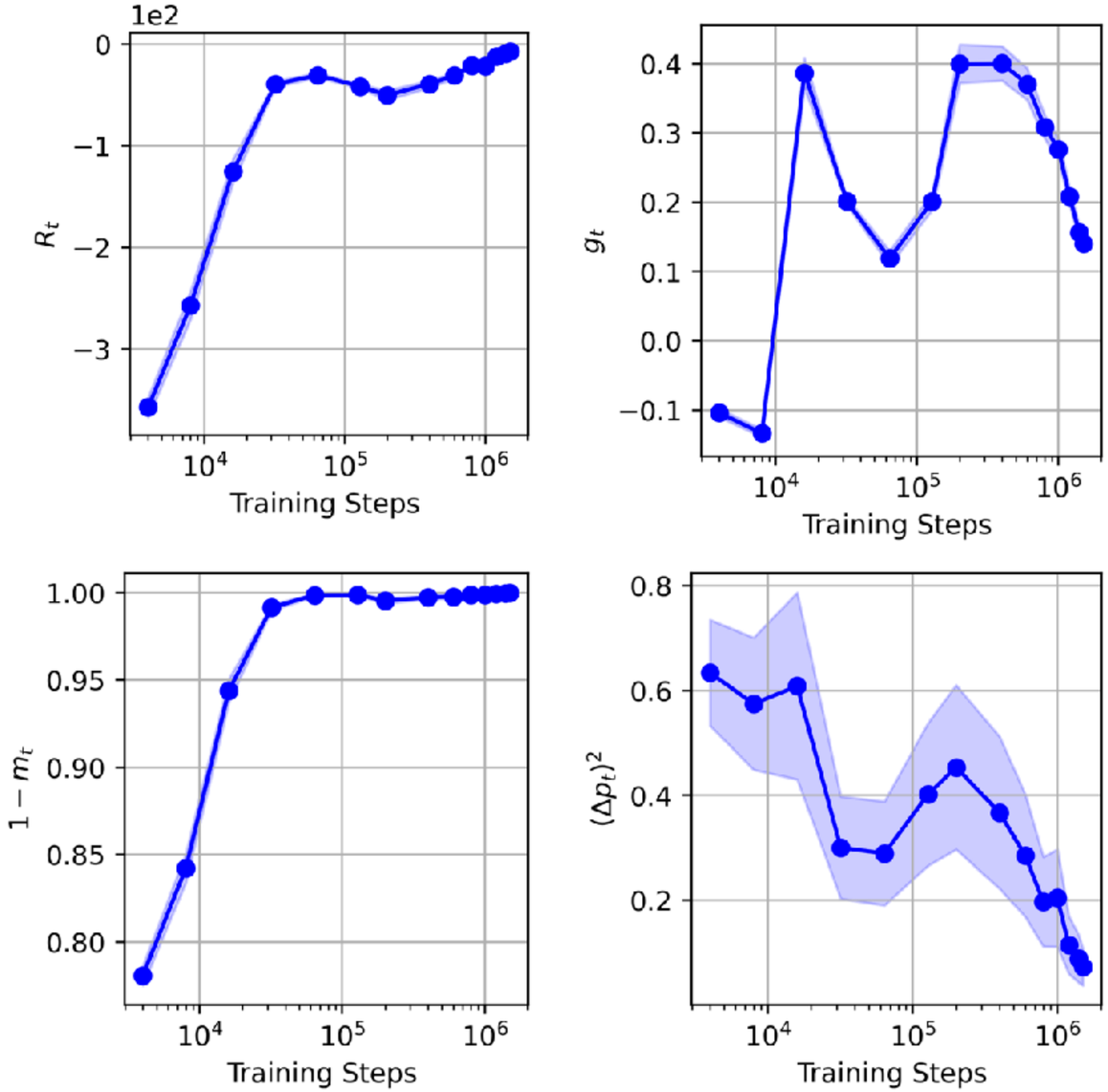
The GasRL simulator: Learning

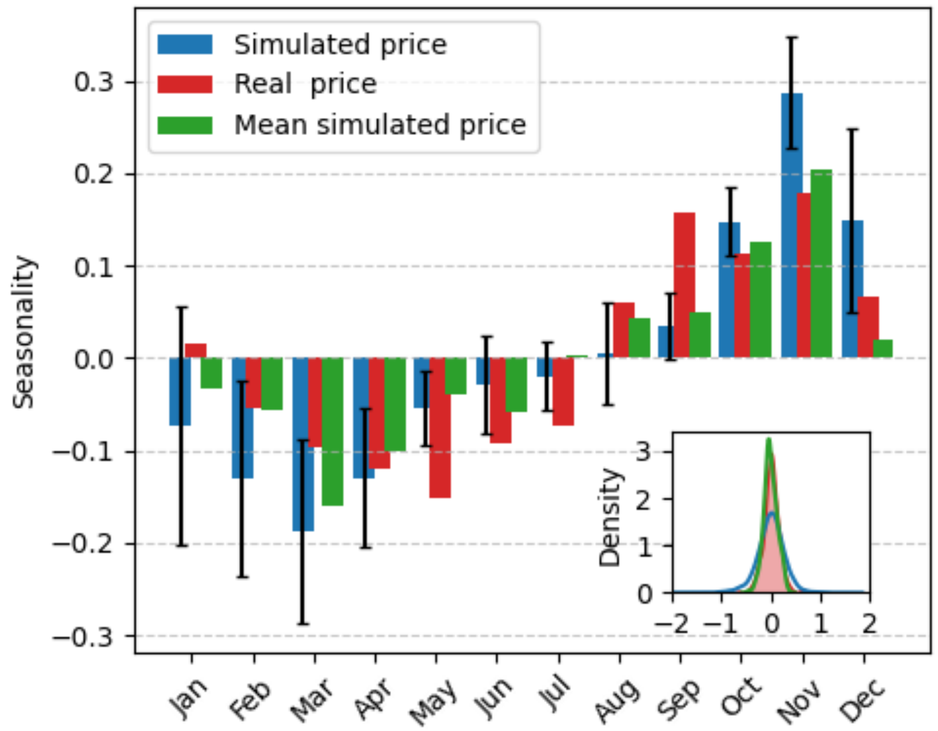


- **SAC** outperform other RL schemes for learning speed and stability
- The RL operator quickly learns a **basic, non profitable, pricing policy**
- Eventually, the RL operator learns a **sophisticated, realistic-looking, profitable policy**

$\theta_n = 0$

The GasRL simulator: Validation



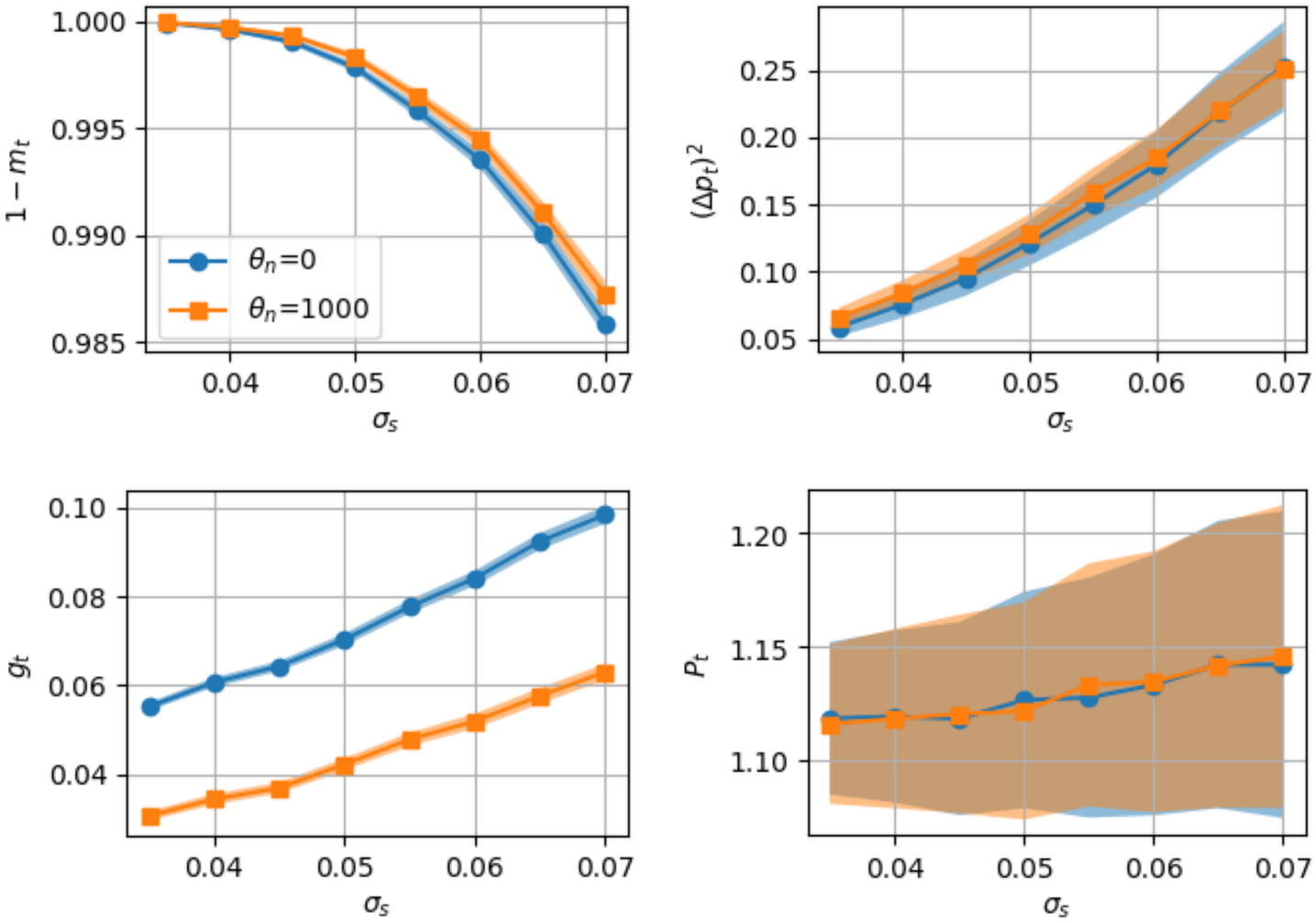


- **During training**, reward and market success rise, but bank account is sometimes lost in favour of a lower volatility.
- Basic features of real prices such as **seasonality and volatility are reproduced,** in spite of not being present in the training process

$$R_t = \Delta g_t - \theta_v (\Delta p_t)^2 - \theta_m m_t (1 + \tilde{m}_t)$$

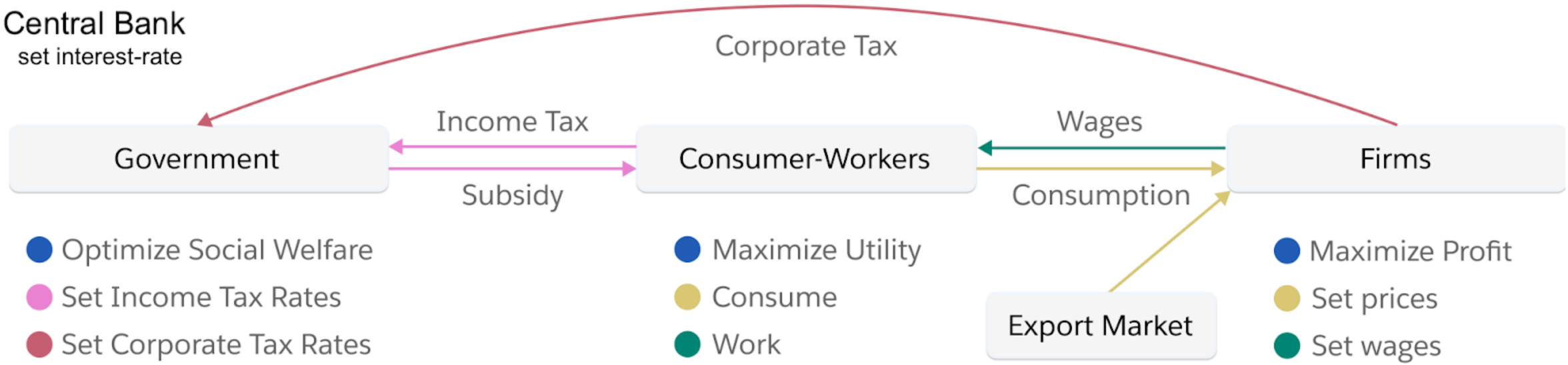
- $\theta_n = 0$

The GasRL simulator: A policy experiment



- We train with the model with (***θ_n =1000***) or without (***θ_n =0***) regulatory constraint and check the performance for **test-only increases in supply shock volatility**
- The constraint slightly **improves market success robustness** …
- … **at the expense of** reduced profits and slightly increased price volatility, but with no effect on overall prices

$$R_t = \underbrace{\Delta g_t}_{\text{profits}} - \theta_v \underbrace{(\Delta p_t)^2}_{\text{price volatility}} - \theta_m \underbrace{m_t(1+\tilde{m}_t)}_{\text{market failure}} - \theta_n \underbrace{n_t(1+\tilde{n}_t)}_{\text{regulatory compliance}}$$

T. Bacaloni, A. Glielmo, M. Taboga
Natural-gas storage modelling by deep reinforcement learning, ***ICAIF 2025***

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